

Part 5 – Train Your First Model

1. What is the slope?

The slope is the coefficient value of each feature in Linear Regression. It shows how much the predicted marks change when a feature increases by one unit. Each feature has its own slope value.

2. What is the intercept?

The intercept is the starting value of the prediction when all feature values are zero. It acts as the base value of the regression equation before the effect of the features is added.

3. What do they mean?

The slope and intercept together form the prediction equation. The slope explains the impact of each feature on the output, while the intercept provides the starting point. They help the model make predictions.

Part 8 – Model Evaluation

1. What is MAE?

Mean Absolute Error (MAE) measures the average difference between actual values and predicted values. A lower MAE means the predictions are closer to the real values. It is used to evaluate model accuracy.

2. What is R² Score?

R² Score shows how well the model explains the relationship between features and the target variable. Its value ranges from 0 to 1. A value closer to 1 indicates better model performance.

3. Is your model performing well?

If the MAE is low and the R² Score is close to 1, the model is performing well. This means the predictions are accurate and the model has learned the pattern in the dataset effectively.

Part 10 – Student Research Activity

1. What is Machine Learning?

Machine Learning is a branch of artificial intelligence that enables computers to learn from data and make decisions without being explicitly programmed. It identifies patterns in data and uses them to make predictions. Machine Learning is used in many fields such as healthcare, banking, education, and online shopping. It helps automate tasks, improve accuracy, and solve real-world problems efficiently. The more relevant data a model receives, the better it can learn and perform.

2. What is Linear Regression?

Linear Regression is a Machine Learning algorithm used to predict numerical values. It finds the relationship between input features and an output variable by fitting a straight line through the data. It is simple, easy to understand, and commonly used for prediction tasks. Examples include predicting student marks, salaries, and house prices. Linear Regression is often the first algorithm learned by beginners in Machine Learning.

3. What are Features?

Features are the input variables used by a Machine Learning model to make predictions. In a student dataset, study hours, attendance, and assignments are examples of features. They provide information that helps the model understand patterns in the data. Choosing relevant features improves model accuracy and performance. Features are important because they directly influence the prediction made by the model.

4. What are Labels?

Labels are the target values that a Machine Learning model tries to predict. In the student performance dataset, marks are the label because they are the output being predicted. Labels provide the correct answers during training. The model learns the relationship between features and labels. Accurate labels help the model learn effectively and make better predictions on new data.

5. Difference Between Training Data and Testing Data

Training data is used to teach the Machine Learning model. The model learns patterns and relationships from this data. Testing data is used after training to evaluate the model's performance on unseen data. Training data helps the model learn, while testing data helps measure accuracy. A common split is 80% training data and 20% testing data. Both are necessary for building reliable models.

6. Real-Life Applications of Machine Learning

Machine Learning is used in many real-world applications. Online shopping websites recommend products based on customer interests. Banks use it to detect fraud and secure transactions. Healthcare systems use it to assist in disease diagnosis. Social media platforms use it for content recommendations. Navigation apps predict traffic and suggest routes. Smartphones use Machine Learning for voice assistants, face recognition, and photo organization.